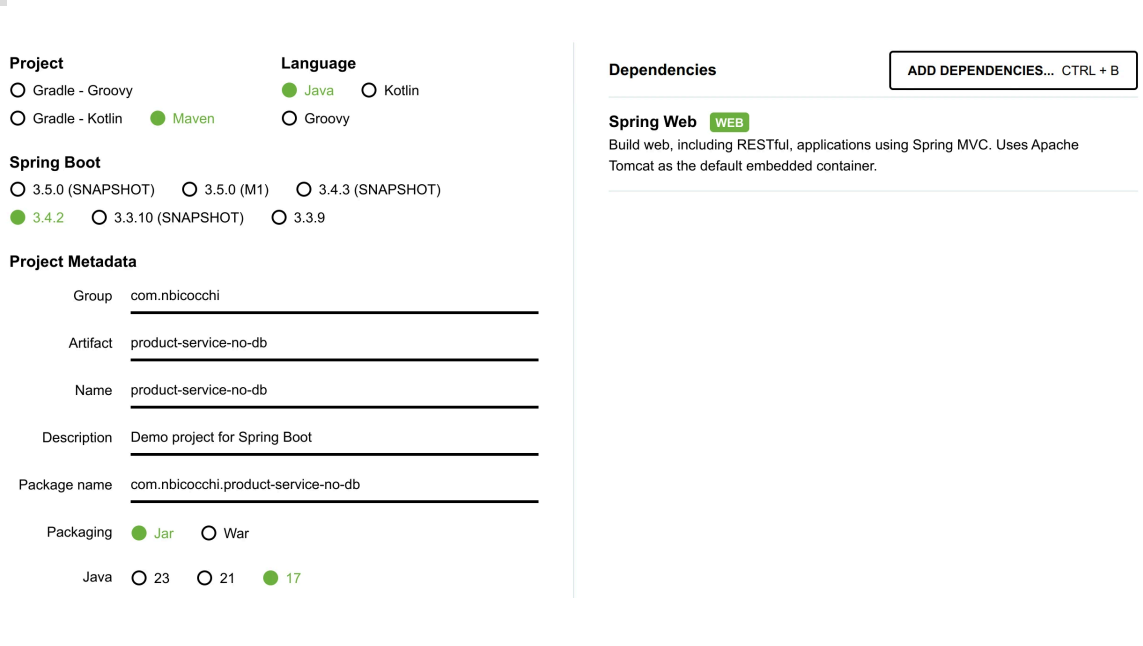
Real-World Project

Create Project

1. Go to Spring Initializr.
2. Select **Maven Project**, **Java**, and **Spring Boot version** (latest stable).
3. Fill in the project metadata:
    - **Group:** `com.nbicocchi`
    - **Artifact:** `product-service-no-db`
4. Add the dependency **Spring Web**.
5. Click **Generate** to download the project as a `.zip`.
6. Extract and open the project in your IDE (e.g., IntelliJ IDEA, VS Code).

> ***Note:*** *Spring Web keeps the app running and allows us to expose endpoints later.*



Project
○ Gradle - Groovy
○ Gradle - Kotlin
● Maven

Language
● Java
○ Kotlin
○ Groovy

Spring Boot
○ 3.5.0 (SNAPSHOT)
○ 3.5.0 (M1)
○ 3.4.3 (SNAPSHOT)
● 3.4.2
○ 3.3.10 (SNAPSHOT)
○ 3.3.9

Project Metadata

Group: com.nbicocchi
Artifact: product-service-no-db
Name: product-service-no-db
Description: Demo project for Spring Boot
Package name: com.nbicocchi.product-service-no-db
Packaging: ● Jar ○ War
Java: ○ 23 ○ 21 ● 17

Dependencies

Spring Web WEB
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

Persistence Layer

We'll create a simple persistence layer under the package *com.nbicocchi.product.persistence.model*, by adding a *Product* class.

```
@AllArgsConstructor
@NoArgsConstructor
@RequiredArgsConstructor
@Data
@Entity
public class Product {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    @NonNull @EqualsAndHashCode.Include private String uuid;
    @NonNull private String name;
    @NonNull private Double weight;
}
```

This class has 3 key attributes: *uuid*, *name* and *weight*.

Next, we'll add the repository under the package *com.nbicocchi.product.persistence.repository*:

```
@Repository
public interface ProductRepository extends CrudRepository<Product, Long> {
    Optional<Product> findByUuid(String name);
}
```

Service Layer

Moving on to the service layer, we'll add a similar service under the package *com.nbicocchi.product.service*:

```
@Service
public class ProductService {
    private final ProductRepository productRepository;

    public ProductService(ProductRepository productRepository) {
        this.productRepository = productRepository;
    }

    public Optional<Product> findByUuid(String uuid) {
        return productRepository.findByUuid(uuid);
    }

    public Iterable<Product> findAll() {
        return productRepository.findAll();
    }

    public Product save(Product product) {
        return productRepository.save(product);
    }

    public void delete(Product product) {
        productRepository.delete(product);
    }
}
```

Presentation Layer

Moving on to the presentation layer, we'll add a similar service interface under the package *com.nbicocchi.controller*:

```
@RestController
@RequestMapping("/products")
public class ProductController {
    ProductService productService;

    public ProductController(ProductService productService) {
        this.productService = productService;
    }

    @GetMapping("/{uuid}")
    public Product findByUuid(@PathVariable String uuid) {
        return productService.findByUuid(uuid).orElseThrow(() -> new
ResponseStatusException(HttpStatus.NOT_FOUND));
    }

    @GetMapping
    public Iterable<Product> findAll() {
        return productService.findAll();
    }

    @PostMapping
    public Product create(@RequestBody Product product) {
        return productService.save(product);
    }

    @PutMapping("/{uuid}")
    public Product update(@PathVariable String uuid, @RequestBody Product product) {
        Optional<Product> optionalProject = productService.findByUuid(uuid);
        optionalProject.orElseThrow(() -> new ResponseStatusException(HttpStatus.NOT_FOUND));
        product.setId(optionalProject.get().getId());
        return productService.save(product);
    }

    @DeleteMapping("/{uuid}")
    public void delete(@PathVariable String uuid) {
        Optional<Product> optionalProject = productService.findByUuid(uuid);
        optionalProject.orElseThrow(() -> new ResponseStatusException(HttpStatus.NOT_FOUND));
        productService.delete(optionalProject.get());
    }
}
```

ApplicationRunner and *CommandLineRunner* Interfaces

In Spring Boot, *CommandLineRunner* and *ApplicationRunner* are two interfaces that allow you to execute code when a Spring Boot application starts. They are typically used to perform some initialization or setup tasks before the application starts processing requests. Both interfaces have a single run method that you need to implement.

```
@Log
@SpringBootApplication
public class App implements ApplicationRunner {
    ProductRepository productRepository;

    public App(ProductRepository productRepository) {
        this.productRepository = productRepository;
    }

    public static void main(final String... args) {
        SpringApplication.run(App.class, args);
    }

    @Override
    public void run(ApplicationArguments args) {
        productRepository.save(new Product("171f5df0-b213-4a40-8ae6-fe82239ab660", "Laptop", 2.2));
        productRepository.save(new Product("f89b6577-3705-414f-8b01-41c091abb5e0", "Bike", 5.5));
        productRepository.save(new Product("b1f4748a-f3cd-4fc3-be58-38316afe1574", "Shirt", 0.2));

        Iterable<Product> products = productRepository.findAll();
        for (Product product : products) {
            log.info(product.toString());
        }
    }
}
```

Testing the Product microservice

See all products

```
curl -X GET http://localhost:7001/products | jq
```

```
[
  {
    "id": 163814,
    "uuid": "171f5df0-b213-4a40-8ae6-fe82239ab660",
    "name": "Laptop",
    "weight": 2.2
  },
  {
    "id": 269752,
    "uuid": "f89b6577-3705-414f-8b01-41c091abb5e0",
    "name": "Bike",
    "weight": 5.5
  },
  {
    "id": 328487,
    "uuid": "b1f4748a-f3cd-4fc3-be58-38316afe1574",
    "name": "Shirt",
    "weight": 0.2
  }
]
```

See one product

```
curl -X GET http://localhost:7001/products/171f5df0-b213-4a40-8ae6-fe82239ab660 | jq
```

```
{
  "id": 833124,
  "uuid": "171f5df0-b213-4a40-8ae6-fe82239ab660",
  "name": "Laptop",
  "weight": 2.2
}
```

Add a product

```
curl -X POST http://localhost:7001/products -H "Content-Type: application/json" -d '{
    "uuid": "b1f4748a-0000-4fc3-be58-38316afe1574",
    "name": "Puppet",
    "weight": 0.2
  }'
```

```
{"id":777523,"uuid":"b1f4748a-0000-4fc3-be58-38316afe1574","name":"Puppet","weight":0.2}
```

Update a product

```
curl -X PUT http://localhost:7001/products/b1f4748a-0000-4fc3-be58-38316afe1574 -H "Content-Type:
application/json" -d '{
    "uuid": "b1f4748a-0000-4fc3-be58-38316afe1574",
    "name": "Puppet, but nicer",
    "weight": 0.3
  }'
```

```
{"id":777523,"uuid":"b1f4748a-0000-4fc3-be58-38316afe1574","name":"Puppet, but nicer","weight":0.3}
```

Delete a product

```
curl -X DELETE http://localhost:7001/products/b1f4748a-f3cd-4fc3-be58-38316afe1574 | jq
```

Resources